

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2025

Programme: BE

Full Marks : 100

Course: Programming in C (New)

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the different types of programming languages with examples. 8
- b) Draw a flowchart to calculate the sum of the first N natural numbers, where N is entered by the user. 7
2. a) What is a variable in C? What are the rules for naming variables. 7
Mention appropriate data type for storing following information:
 - i. Distance jumped by athlete
 - ii. Prime number between 5 to 500
 - iii. Age of a student
 - iv. Address of an employee
- b) Why do you use break and continue statement in your program? 8
Explain with suitable example program.
3. a) WAP to print the following pattern: 8
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

OR

WAP to check whether an integer entered by user is palindrome or not.

- b) Write a program to input a 3×3 matrix and find the sum of each row and display them. 7
4. a) What are the string handling functions in C? Write a program to input a string and display its reverse using standard library function. 8

- b) Explain importance of recursion. Write a function in C that calculates the factorial of a number. 7

OR

Explain global and local variable in C with an example.

5. a) What is recursion? Compare and contrast recursion and iteration with examples. 7
- b) Write a program to show how pointers can be used to access array elements. 8
6. a) Given a structure: 8

Name	Address	Telephone	Salary	Date of birth		
				mm	dd	yy

Write a program to input data of n employee and display records of those employees living in "Dhangadhi" and whose salary is above 10000.

- b) Explain the different file opening modes with examples. 7
7. Write short notes on: (**Any two**) 2×5
- a) Formatted I/O functions
- b) Function call by value
- c) Dynamic memory allocation